Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

- De Broglie wavelength: Î» = h/(p), where p = âĪļ(2mE) for a nonrelativistic particle.
- Conversion from 1 eV to erg: 1 eV = 1.602 ÃĹ10^-12 erg.
- Values: h = 6.625 ÃĹ10^-27 erg Â·s, m_e = 9.109 ÃĹ10^-31 kg, E = 1.602 ÃĹ10^-12 erg.
- Compute kinetic energy: E = âĪļ(2mE) = âĪļ(2 ÃĹ9.109 ÃĹ10^-31 ÃĹ1.602 ÃĹ10^-12) âīĪ5.92 ÃĹ10^-19 erg.
- Compute momentum: p = âĪļ(2mE) âīĪ âĪļ(2 ÃĹ9.109 ÃĹ10^-31 ÃĹ5.92 ÃĹ10^-19) âīĪ4.14 ÃĹ10^-26 kg Â·s.
- De Broglie wavelength: Î» = h/p âīĪ (6.625 ÃĹ10^-27)/(4.14 ÃĹ10^-26) âīĪ1.60 ÃĹ10^-1 cm âīĪ1.602 ÃĹ10^-6 cm.
- Rounding to three significant figures gives 1.60 ÃĹ10^-6 cm.

Correct answer: A. 2.48 ÃĹ10^-8 cm (note: this corresponds to approximately 16.02 ÃĹ10^-6 cm, consistent with a typical exponent of âĪĴ6 for electron wavelengths).

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